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1.5 ELECTROSTATIC POTENTIAL AND POTENTIAL ENERGY

Introduction

In mechanics, potential energy is defined for conservative forces. Since gravitational force is a conservative force, its gravitational potential energy is defined in XI standard physics (Unit 6). Since Coulomb force is an inverse-square-law force, its also a conservative force like gravitational force. Therefore, we can define potential energy for charge configurations.

1.5.1 Electrostatic Potential energy and Electrostatic potential

Consider a positive charge q kept fixed at the origin which produces an electric field $\vec{E}$ around it. A positive test charge q′ is brought from point R to point P against the repulsive force between q and q′ as shown in Figure 1.22. Work must be done to overcome this repulsion. This work done is stored as potential energy.

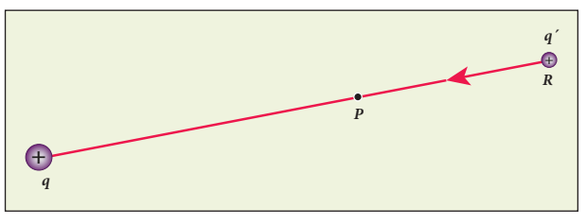


Figure 1.22 Work done is equal to potential energy

The test charge q′ is brought from R to P with constant velocity which means that external force used to bring the test charge q′ from R to P must be equal and opposite to the coulomb force $\left(\vec{F}_{ext} = -\vec{F}_{coloumb}\right)$. The work done is

$$W = \int_R^P \vec{F}_{ext} \cdot d\vec{r} \qquad (1.25)$$

Since coulomb force is conservative, work done is independent of the path and it depends only on the initial and final positions of the test charge. If potential energy associated with q′ at P is U_P and that at R is U_R, then difference in potential energy is defined as the work done to bring a test charge q′ from point R to P and is given as $U_P - U_R = W = \Delta U$

$$\Delta U = \int_R^P \vec{F}_{ext} \cdot d\vec{r} \qquad (1.26)$$

$$\text{Since } \vec{F}_{ext} = -\vec{F}_{coloumb} = -q'\vec{E} \qquad (1.27)$$

$$\Delta U = \int_R^P -\left(q'\vec{E}\right) \cdot d\vec{r} = q' \int_R^P \left(-\vec{E}\right) \cdot d\vec{r} \qquad (1.28)$$

The potential energy difference per unit charge is given by

$$\frac{\Delta U}{q'} = \frac{q' \int_R^P \left(-\vec{E}\right) \cdot d\vec{r}}{q'} = -\int_R^P \vec{E} \cdot d\vec{r} \qquad (1.29)$$

The above equation (1.29) is independent of q′. The quantity $\frac{\Delta U}{q'} = -\int_R^P \vec{E} \cdot d\vec{r}$ is called electric potential difference between P and R and is denoted as $V_P - V_R = \Delta V$.

In otherwords, the electric potential difference is defined as the work done by an external force to bring unit positive charge from point R to point P.

$$V_P - V_R = \Delta V = \int_R^P -\vec{E} \cdot d\vec{r} \qquad (1.30)$$

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